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SINGLE GATE VALVE

Abstract

A valve has a body having a through bore. A gate is disposed in a passage transverse to the through bore, with the gate configured for motion between a position with an opening on the gate coincident with the through bore to permit fluid flow via the through bore and a position to restrict fluid flow via the through bore. The gate is configured for motion along the transverse passage in one direction in response to a first force acting on the gate and in another direction in response to a second force acting on the gate. A method of operating a valve.

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Background/Summary

BACKGROUND

[0001] This disclosure relates to the field of mechanical valves. More specifically, the disclosure relates to gate valve designs for selective fluid communication in general applications.

[0002] Valves to control the transmission and flow of fluids have been in use for centuries. Gate valves are well known and applied in various industries. In oilfield operations (e.g., fracking applications), gate valves are commonly used to handle fluid flow at each well. Such valves are exposed to extremely harsh fluids (e.g., sand slurries) that significantly reduce the operational life of the gates. Conventional gate valves are designed with a stem and stem packing configuration, which is less than ideal for such applications. A typical gate valve applied in a fracking operation will require refurbishment approximately every two weeks during operation. This type of maintenance is time consuming and costly, often requiring shipment of the valve to a workshop for repair. Thus, a need remains for improved valve designs to handle various types of fluid transmission.

SUMMARY

[0003] One aspect of the present disclosure is a valve including a main body having a through bore, a passage transverse to the through bore, and a gate disposed in the transverse passage. The gate is configured for motion between a position with an opening on the gate coincident with the through bore to permit fluid flow via the through bore and a position to restrict fluid flow via the through bore. The gate is configured for motion along the transverse passage in one direction in response to a first force acting on the gate and in another direction in response to a second force acting on the gate.

[0004] Another aspect of the present disclosure is a method of operating a valve having a main body with a through bore. The method includes applying a force on a gate to move the gate along a passage in the main body transverse to the through bore, wherein the gate is configured for motion along the transverse passage between a position with an opening on the gate coincident with the through bore to permit fluid flow via the through bore and a position to restrict fluid flow via the through bore. The gate is configured for motion along the transverse passage in one direction in response to a first force acting on the gate and in another direction in response to a second force acting on the gate.

Description

DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** shows an oblique view of a valve embodiment according to this disclosure.

[0006] FIG. **2** shows a view of a gate cartridge in a valve embodiment according to this disclosure.

[0007] FIG. **3**A shows a schematic of a valve embodiment in an open position according to this disclosure.

[0008] FIG. **3**B shows the valve of FIG. **3**A in a closed position according to this disclosure.

[0009] FIG. **4** shows an oblique transparency of valve embodiment according to this disclosure.

[0010] FIG. **5** shows a cross section view of a valve embodiment according to this disclosure.

[0011] FIG. **6** shows an overhead cutaway view of a valve embodiment according to this

disclosure.
[0012] FIG. **7** shows a schematic of a valve spool embodiment according to this disclosure.
[0013] FIG. **8** shows a schematic of a valve embodiment with a removable end cap according to this disclosure.
[0014] FIG. **9** shows a valve embodiment disposed in a fluid system according to this disclosure.
[0015] FIG. **10** shows a schematic of another valve embodiment according to this disclosure.
[0016] FIG. **11** shows a cross section view of a valve embodiment according to this disclosure.
[0017] FIG. **12** shows the valve of FIG. **11** in a closed position according to this disclosure.
[0018] FIG. **13** shows a fluid/gas flow schematic of a valve embodiment according to this disclosure.
[0019] FIG. **14** shows a cross section view of another valve embodiment according to this disclosure.
[0020] FIG. **15** shows an overhead view of the valve of FIG. **14** according to this disclosure.
[0021] FIG. **16** shows another overhead view of the valve of FIG. **14** according to this disclosure.

DETAILED DESCRIPTION

[0022] Illustrative embodiments are disclosed herein. In the interest of clarity, not all features of an actual implementation may be described. In the development of any such actual embodiment, numerous implementation-specific decisions may need to be made to achieve the design-specific goals, which may vary from one implementation to another. It will be appreciated that such a development effort, while possibly complex and time-consuming, would nevertheless be a routine undertaking for persons of ordinary skill in the art having the benefit of this disclosure. The disclosed embodiments are not to be limited to the precise arrangements and configurations shown in the figures, in which like reference numerals may identify like elements. Also, the figures are not necessarily drawn to scale, and certain features may be shown exaggerated in scale or in generalized or schematic form, in the interest of clarity and conciseness.
[0023] FIG. **1** shows a valve **10** embodiment according to this disclosure. The valve **10** has a main body **12**, a first end **14**, a second end **16**, a first surface **18**, and a second surface **20** opposite the first surface. A through bore **22** traverses through the body **12**, providing an open passage between the first surface **18** and the second surface **20**. Although the embodiment of FIG. **1** shows the through bore **22** configured as a cylindrical opening, the bore may be configured in other geometrical shapes as desired for the application (e.g., oval, octagonal, etc.). The body **12** may be formed of any suitable material depending on the application (e.g., metal, composites, plastics, synthetic materials, etc.). Although FIG. **1** shows an embodiment configured with a generally planar body **12** design, embodiments may be implemented with bodies comprising other geometrical designs more suitable for the desired application.
[0024] Some valve **10** embodiments may be configured with threaded holes **24** formed on each surface **18**, **20** to receive mounting bolts for mounting of the valve **10** onto a fluid transmission system as known in the art (see FIG. **9**). Embodiments may also be configured with the holes **24** passing through the entire body **12** for engagement of the valve **10** to flange units using extended length bolts. It will be appreciated by those skilled in the art that other embodiments may be configured for disposal of the valve **10** onto fluid lines or systems in various fashions depending on the application (e.g., welded onto a line, affixed with clamps, etc.).
[0025] FIG. **2** shows a view of a valve **10** embodiment according to this disclosure. The valve **10** is shown with a gate **26** extending out of a slot **28** formed within the body **12**. In this embodiment, the gate **26** is configured as a rectangular-shaped planar body **29** with a first face **30** and a second face **32**. The gate **26** has an opening **34** formed near a first end **31**. The opening **34** is preferably shaped to coincide with the geometric shape of the through bore **22** formed in the valve **10** body. The gate **26** provides a solid surface area **36** near a second end **35**. FIG. **2** shows the opening **34** surrounded by a seal trench **38**. The other end with the solid surface area **36** also has a seal trench **40** formed thereon. Although not shown in FIG. **2**, the second face **32** of the gate **26** is configured with

matching seal trenches **38′**, **40′**. In essence, the first face **30** of the gate **26** is a mirror image of the second face **32**.

[0026] The gate **26** may be formed of any suitable material depending on the application (e.g., metal, composites, plastics, synthetic materials, etc.). As shown in FIG. **2**, the gate **26** is disposed in the slot **28**, which forms a passage transverse to the through bore **22** along the longitudinal axis of the valve **10** body **12** (see FIGS. **4**, **5**, **6**). The gate **26** is formed to fit within the slot **28** at a close tolerance yet allowing the gate to move or slide within the slot as described below. As depicted in FIG. **2**, the gate **26** is configured for easy insertion and extraction from the slot **28**, akin to a cartridge in a player. This provides a notable advantage compared to conventional valve designs, which typically require removal of the entire valve for repair or refurbishment. Some embodiments may be implemented with a seal or band **42** (e.g., a rubber band) disposed in a groove **44**A formed on the gate **26** surface near the center to provide a wiper for the internal surface of the slot **28** as the gate moves back and forth therein. Embodiments may also be implemented with additional wiper bands or seals disposed in other grooves **44**B on the gate **26**. FIG. **2** also shows the gate **26** configured with a threaded port **46**A at one end (further described with respect to FIG. **7**).

[0027] FIG. **3**A shows an overhead cutaway schematic of a valve **10** embodiment of this disclosure. The gate **26** is shown in the open position, with the opening **34** coincident and aligned with the through bore **22**. In this open position, any fluid traversing the through bore **22** from either the first face **30** or the second face **32** is free to flow through the gate **26** opening **34** and in-out through the valve **10**.

[0028] FIG. **3**B shows the valve **10** of FIG. **3**A with the gate **26** in the closed position. When the gate **26** is moved (as described below) to the other end of the slot **28** compared to FIG. **3**A, the solid surface area **36** of the gate **26** fully covers the through bore **22**, preventing fluid passage therethrough.

[0029] Turning to FIG. **4**, a transparency view of a valve **10** embodiment according to this disclosure is shown. The gate **26** is shown in the open position, as described with respect to the embodiment of FIG. **3**A. In this embodiment, the gate **26** is configured with a first seal assembly **48** disposed at the gate opening **34** and a second seal assembly **50** disposed at the solid surface area **36** (see FIG. **2**). First seal assembly **48** includes a first seal **48**A disposed in the seal trench **38** formed on the first face **30** of the gate **26**, and a second seal **48**B disposed in a mirroring seal trench **38′** formed on the second face **32** of the gate (see description of FIG. **2**). The second seal assembly **50** likewise includes a first seal **50**A disposed in the seal trench **40** formed on the first face **30** of the gate **26**, and a second seal **50**B disposed in a mirroring seal trench **40′** formed on the second face **32** of the gate (see description of FIG. **2**).

[0030] FIG. **5** shows a cross section of a gate **10** embodiment according to this disclosure. As depicted in FIG. **5**, the first and second seal assemblies **48**, **50** may be implemented with conventional radial seals **48**A, **48**B, **50**A, **50**B (see FIG. **4**) to restrict fluid passage between the through bore **22** and the slot **28**. Some embodiments may also be implemented with the seals **48**A, **48**B, **50**A, **50**B configured for energization to urge the respective seal faces against the surfaces to be sealed. Such seals are further described in Intl. Pat. Apps. WO20211142004 and WO20211141999, both assigned to the present assignee and incorporated herein by reference.

[0031] FIG. **5** also shows a first fluid port **52** leading to an internal fluid passage **52**A that traverses the valve **10** body **12** to provide a fluid feed to the slot **28** at the first end **14**. A second fluid port **54** is disposed at the second end **16** of the valve **10** and leads to another internal fluid passage **54**A that traverses the valve body **12** to provide a fluid feed to the slot **28** at the second end. As shown in FIG. **5**, the valve **10** is in the open position with the gate **26** abutting the slot **28** wall at the first end **14** of the valve. As described herein, in this position the gate **26** opening **34** is coincident with the through bore **22**, permitting fluid flow therethrough from either end across the valve **10** body **12**. The gate **26** is set in the open position by maintaining a constant fluid pressure in the slot **28** at the second end **16** via fluid passage **54**A, as depicted by the arrow in FIG. **5**.

[0032] To close the valve **10**, fluid is introduced under pressure through the first fluid port **52**, via fluid passage **52**A and into the slot **28** at the first valve **10** end **14**. Simultaneously, fluid pressure is released from the slot **28** at the second end **16** via second fluid port **54**. As fluid pressure at the first end **14** overcomes the pressure at the second end **16** of the slot **28**, the gate **26** is pushed from the open to the closed position (left to right in the page). FIG. **3**B shows a valve **10** with the gate **26** in the closed position. Although the first and second fluid ports **50**, **52** are shown disposed at the first planar surface **18** of the valve **10**, it will be appreciated that other embodiments may be implemented with the ports located at other surfaces (e.g., first end **14**, second end **16**, second planar surface **20**, etc.) to facilitate mounting of the valve depending on the application.
[0033] Fluid pressure to move the gate **26** from the open-to-closed position, and vice-versa, as described herein, may be provided by a separate pump and fluid (e.g., hydraulic fluid) reservoir system coupled to the first and second fluid ports **52**, **54** (sec FIGS. **9** and **13**). As such, the valve **10** embodiments provide a closed system for the fluid to move the gate **26**. FIG. **5** also shows a valve **10** embodiment configured with a first annular seat **56** disposed in a channel **58** formed in the body **12** above the gate **26** coincident with the through bore **22**. A second annular seat **60** is disposed in another channel **62** formed in the body **12** below the gate **26** coincident with the through bore **22**. The first and second annular seats **56**, **60** may be formed of more durable materials (e.g., stainless steel, INCONEL™, ceramics, tungsten carbide, etc.) compared to the valve **10** body **12**. Since scaling at the through bore **22**/gate **26** junction is important to an efficient and effective valve **10**, the annular seats **56**, **60** provide a hardwearing corrosive-resistant surface to sustain a good seal via the seal assemblies **48**, **50**, particularly when implemented with energizable seals. In applications with fluids containing damaging or abrasive elements (e.g., sand contamination), the combination of seals **48**, **50** and seats **56**, **60** provides the necessary scaling to prevent migration of undesired contaminants internally within the valve **10**. Placement of the seal assemblies **48**, **50** on the gate **26** cartridge provides a significant advantage compared to conventional valves. Valves **10** configured with hardwearing annular seats **56**, **60** can provide the required sealing ability for extended operation since the seal assemblies **48**, **50** can be easily replaced in the field (without having to remove the valve **10** from the system) by extraction and refurbishment or replacement of the gate **26** cartridge as described herein.
[0034] Turning to FIG. **6**, a transparency cutaway view of the top of a valve **10** embodiment according to this disclosure is shown. Valve **10** embodiments implemented with energizable radial seal assemblies **48**, **50** entail the use of a pressurized fluid, gas, compounds, springs, or combinations of the aforementioned to energize the seals to provide a superior seal at the through bore **22**/gate **26** junction. Usable seal embodiments include those described in Intl. Pat. Apps. WO20211142004 and WO20211141999. FIG. **6** shows a gate **26** embodiment configured with internal fluid passages to channel fluid to energize the seals disposed on the gate.
[0035] The valve **10** of FIG. **6** is shown in the open position, with the gate opening **34** coincident with the through bore **22**. In this mode, the seals **48**A, **48**B of the first seal assembly **48** are energized up by fluid pressure provided by a reservoir of fluid (e.g., hydraulic fluid) fully contained within the gate **26**. When the valve **10** is in the closed position, the second seal assembly **50** (on the left side of the page) is energized up. During the transition phase between open-close-open, the respectively engaged seals begin to de-energize, allowing the gate **26** to move while the other seals become energized.
[0036] The gate **26** is configured with two fluid timing circuits. Each circuit activates one of the seal assemblies **48**, **50**. Each circuit is implemented by a valve spool and a fluid reservoir interconnected via internal fluid passages disposed on the gate **26**. FIG. **7** shows a cross section of a valve spool **64** embodiment according to this disclosure. The spool **64** is configured with a plug **66** that is threaded into an internal passage **68** within the gate **26** (e.g., via port **46**A of FIG. **2**). An internal plunger **70** provides an annular flow space **72** between sealed ends (e.g., via O-rings). When acted upon by fluid pressure or physical contact with the gate **26** body, the plunger **70** is free

to move within the internal passage **68** in the gate. In one position, a distal end **74** of the plunger extends outward past the end of the gate **26** (see plunger **94** of FIG. **6**).

[0037] Returning to FIG. **6**, the valve **10** is in the open position, allowing unrestricted fluid passage through the gate **26** via the opening **34**/through bore **22**. In this open position, a first fluid reservoir **76** in the gate **26** provides hydraulic fluid under pressure (via a spring-piston unit) through a first internal fluid passage **78** that provides an outlet **80** at the seal trenches **38**, **38**′ (see FIG. **2**) underneath each seal **48**A, **48**B to energize up each seal. In this position, the second end **35** of the gate **26** is pushed up against the valve body **12** wall by the fluid pressure applied to the first end **31** of the gate via internal fluid passage **54**A. The distal end of a first valve spool **82** plunger is pressed into the receiving port **84** as the gate **26** abuts the body **12** wall. With the plunger in this position, the annular flow space provided by the plunger links the internal fluid passages as shown to allow fluid under pressure from the first fluid reservoir **76** to flow through the outlet **80** to energize up the seals **48**A, **48**B against the surfaces of the respective first and second annular seats **56**, **60** (see FIG. **5**).

[0038] The plunger on a second valve spool **86** is positioned to block fluid flow via the respective internal fluid passages leading to another outlet **88** at the seal trenches **40**, **40**′ of the other seals **50**A, **50**B (see FIG. **5**) to keep those seals de-energized. When activating the gate **26** from the open position (as shown in FIG. **6**) to the closed position by activation fluid pressure via internal fluid passage **52**A, a second fluid reservoir **90** in the gate **26** commences drawing in the fluid through the internal passages **91** (via a spring-piston unit) to allow the fluid under the seals **48**A, **48**B to discharge into the reservoir, allowing the seals to de-energize and release sealing pressure against the annular seat **56**, **60** surfaces. Simultaneously, as the gate **26** transitions to the closed position (left to right in the page), the other seals **50**A, **50**B begin to energize up via fluid pressure through the internal passages as the gate **26** moves to the closed position. When the gate **26** is in the fully closed position, with the first end **31** of the gate abutting against the valve body **12** wall (right side of FIG. **6**), the extended distal end **94** of the second valve spool **86** will be pressed into the receiving port to allow maximum fluid flow under pressure from the second fluid reservoir **90** to flow through the outlet **88** to energize up the seals **50**A, **50**B against the surfaces of the annular seat **56**, **60**, while the other seals **48**A, **48**B are de-energized.

[0039] The disclosed fluid timing circuits channel the internal gate **26** fluids in this manner as the valve **10** cycles through open-closed sequences. The closed fluid system providing the fluid pressure to move the gate **26** back and forth via ports **52**, **54** and the self-contained gate **26** fluid timing circuits in essence comprise a hydraulics-over-hydraulics closed system, which aids in keeping the fluids free of contaminants.

[0040] By maintaining a good seal while the valve **10** is set in the open or closed position and while the gate **26** is moving, maximum protection is provided against contaminant migration as the valve transitions. For example, when flowing fluids with high sand concentrations, the timed energization of the seal assemblies **48**, **50** keeps the sand in the through bore **22** from migrating into the valve body **12**. By preventing such ingress of debris into the valve **10** body the effective operational life of the valve is extended.

[0041] FIG. **8** shows a blowup of a valve **10** with a detachable end cap **96** embodiment according to this disclosure. As used herein for purposes of this disclosure, the words “detachment”, “detached”, and “detachable” are meant to encompass a component fully separated from other components as well as a component partially separated from other components (e.g., a hinged lid, hinged door, etc.). The valve **10** body **12** embodiment of FIG. **8** shows the body end configured with a seal groove **98** to receive a radial seal **100** (e.g., O-Ring) to maintain a sealed passage **28** for the gate **26**. One or more guide pins **102** may also be inserted in holes **104** formed in the end cap **96** and the valve **10** body **12** to provide increased structural stability. Valve **10** embodiments may be implemented with one detachable end cap **96** disposed solely at one end thereof or with a pair of end caps, each mounted at an opposing end of the valve body. Embodiments implemented with a pair of end caps **96** facilitate the removal and insertion of the gate **26** from either end of the valve **10** body **12**.

[0042] As shown in FIG. **8**, an embodiment may be implemented with H-channels **106** formed transversely to the longitudinal axis of the body **12** on each end of the body and the end cap **96**. When the end cap **96** is fitted against the valve **10** body, a pair of elongated I-bars **108**A, **108**B are used to maintain the cap in place. Each I-bar **108**A, **108**B is inserted from one side of the body **12** to slide into place for complementary engagement within the H-channels **106** formed on the cap **96** and body **12**. The I-bars **108**A, **108**B may be secured in place by a conventional fastener. When it is desired to replace the gate **26** cartridge (e.g., to replace all the working seals), the I-bars **108**A, **108**B are pulled out of the H-channels **106** to free the end cap **96** for detachment to allow access to the gate via the passage **28**. Once the end cap **96** is detached, the gate **26** cartridge can be removed and replaced while keeping the valve **10** in place. It will be appreciated by those skilled in the art that other detachable end cap **96** configurations may be implemented with the valves **10** using conventional hardware means and components.

[0043] FIG. **9** shows a valve **10** embodiment of this disclosure implemented in a fluid flow system **200** (e.g., an oilfield fracking operation). The fluid supply to move the gate **26** in the valve **10**, as disclosed herein, may be provided via supply lines **202**, **204** in the system **200**. Other embodiments may be implemented with the valve **10** configured with its own independent fluid reservoir and pump unit to provide the required fluid pressure to the gate **26**. Valve **10** embodiments may also be implemented with a conventional controller **206** and software as known in the art to activate the fluid flow to operate the gate **26**. Some embodiments may also be implemented with a conventional antenna for wireless remote valve **10** activation via the controller **206**.

[0044] FIG. **10** shows another valve **10** embodiment according to this disclosure. It will be appreciated by those skilled in the art that the closed-fluid seal energization systems provided by the gate **26** embodiments disclosed herein are not limited to any particular operation or implementation. FIG. **10** shows an embodiment implemented with a conventional stemmed gate valve **10** design. The valve **10** is configured for manual operation to open-close the gate **26** to allow fluid passage via the through bore **22** across a first end **210** and a second end **212**. A handle **214** is coupled to a stem **216**, which is linked through the valve **10** body **12** to the gate **26**. Rotation of the handle **214** causes the gate **26** to move between an open and closed position, selectively controlling fluid flow across the through bore **22**. The gate **26** may be configured with the energizable seals **218** and features as disclosed herein.

[0045] FIG. **11** shows a cross section of another valve **10** embodiment according to this disclosure. This embodiment is like the other embodiments disclosed herein. However, one difference is the application of different gate **26** activation forces. A first fluid port **300** leads to an internal fluid passage to provide a fluid feed to the slot **28** at the first end **14** of the body **12**. When fluid (e.g., hydraulic fluid) is introduced under pressure via the port **300**, the fluid pressure provides the motive force against the end of the gate **26** (depicted by the arrow in FIG. **11**) to push the gate within the transverse passage **28**. As shown in FIG. **11**, the valve **10** is in the open position with the gate **26** abutting the slot **28** wall at the second end **16**. As described herein, in this position the gate **26** opening **34** is coincident with the through bore **22**, permitting fluid flow therethrough from either side across the valve **10** body **12**. The gate **26** is set in the open position by maintaining a constant fluid pressure in the slot **28** at the first end **14** via fluid port **300**. A second fluid port **302** is disposed at the second end **16** of the valve **10** and leads to another internal fluid passage to provide a fluid feed to the slot **28** at the second end. In some embodiments, the second port **302** is linked to a gas source to provide a pressurized gas motive force against the end of the gate **26** (depicted by the arrow in FIG. **12**) to push the gate toward the first end **14** within the transverse passage **28**.

[0046] FIG. **13** shows a schematic of the fluid system for the valve **10** embodiments depicted in FIGS. **11** and **12**. When the valve **10** is in the open position (FIG. **11**), the fluid feed **310** provides the motive force acting against the end of the gate **26** as described above. Fluid feed **310** is coupled to the first fluid port **300** (FIG. **11**). The fluid may be provided via a conventional fluid pump **312** and/or a conventional accumulator **314**. The pump **312** may be used to keep the accumulator **314** charged with fluid to a desired pressure. It will be appreciated by those skilled in the art that other pressurized fluid sources may be used to implement embodiments of this disclosure (e.g., remote fluid reservoirs/pumps, ROVs, etc.). A control valve **316** may be linked into the system to actuate fluid flow into the valve **10** via port **300** (see FIG. **11**) to act on the gate, transitioning the valve to the open position as shown in FIG. **11**. On the other side **16** of the valve **10**, a pressurized gas feed **318** provides the motive force acting against the opposite end of the gate **26**. Gas feed **318** is coupled to the second fluid port **302** (FIG. **11**). A gas accumulator or gas bottle **320** may be linked to the feed **318** to provide the gas supply. Any suitable gas may be used depending on the application (e.g., conventional nonreactive gases). It will be appreciated by those skilled in the art that other pressurized gas sources may be used to implement embodiments of this disclosure (e.g., coupled remote gas reservoirs, ROVs, etc.). The fluid system also incorporates a vent valve **322** linked into the line on the first side **14** of the valve **10**. Vented fluid may be diverted to a holding tank/reservoir coupled to the system.

[0047] Returning to FIG. **11**, operation of the fluid system is described. When it is desired to actuate the valve **10** to the open position, fluid is introduced under pressure via port **300**. The fluid is introduced with sufficient pressure to act against the end of the gate **26** (arrow in FIG. **11**), overcoming the force of the gas acting on the opposite side of the gate. The gas on the opposite side of the gate is compressed and migrates to the accumulator or gas bottle **320**. The valve **10** will remain in the open position provided the fluid feed **310** (see FIG. **13**) maintains sufficient pressure against the gate **26** to overcome the gas feed **318** pressure on the opposite side of the gate.

[0048] FIG. **12** shows the valve **10** in the closed position. When it is desired to close the valve **10**, the control valve **316** (see FIG. **13**) is closed to stop fluid ingress via port **300** and vent valve **322** is opened. Once the vent valve **322** is opened, the fluid pressure on the gate **26** end drops very quickly. As the fluid pressure drops on the first end **14** of the gate **26**, the compressed gas expands in the closed chamber supplied by the accumulator or gas bottle **320**. As the gas expands, the pressure acts of the end of the gate **26** (arrow in FIG. **12**) to shift the gate into the closed position. It will be appreciated that embodiments as disclosed in FIGS. **11** and **12** provide a fail-safe valve **10** whereupon the gate **26** automatically reverts to the closed position (via expanding pressurized gas feed **318**) when fluid pressure drops or ceases via fluid feed **310**. Embodiments may be implemented wherein the control valve **316** and/or the vent valve **322** (FIG. **13**) may be actuated manually, remotely, and/or automatically and autonomously.

[0049] As with other embodiments of this disclosure, the embodiments of FIGS. **11** and **12** may be implemented with first and second annular seats **56**, **60** disposed in the main body **12** as previously described. In Some embodiments, the annular seats **56**, **60** may be configured with one or more radial/face seals **324** to provide greater sealing integrity. The valves **10** may also be configured with first and second seal assemblies **48**, **50**, which can be implemented with conventional radial seals **48**A, **48**B, **50**A, **50**B (see FIG. **4**) to restrict fluid passage between the through bore **22** and the slot **28**. Some embodiments may also be implemented with the seals **48**A, **48**B, **50**A, **50**B configured for energization to urge the respective seal faces against the surfaces to be sealed. Such seals are further described in Intl. Pat. Apps. WO20211142004 and WO20211141999. In some embodiments, the valves **10** may be configured for seal **48**A, **48**B, **50**A, **50**B energization via the expanding gas from gas feed **318** as described above.

[0050] As shown in FIGS. **11** and **12**, some valve **10** embodiments may be implemented with elongated saw-tooth bars **326** that slide into complementary channels **328** formed transversely to the longitudinal axis of the body **12** on each end of the body and the end caps **96**. When the end cap **96** is fitted against the valve **10** body, a pair of elongated saw-tooth bars **326** are used to maintain the cap in place. The saw-tooth bars **326** may be secured in place by a conventional fastener. When it is desired to replace the gate **26** cartridge (e.g., to replace all the working seals), the saw-tooth bars **326** are pulled out of the channels **328** to free the end cap **96** for detachment to allow access to the gate via the passage **28**. Once an end cap **96** is detached, the gate **26** cartridge and/or the annular seats **56**, **60** can be removed and replaced while keeping the valve **10** mounted in place.

[0051] FIG. **14** shows a cross section of another valve **10** embodiment according to this disclosure. This embodiment incorporates an annular insert **332** disposed within the main body **12**, with a sleeve portion fitted coincident with the through bore **22**. The insert **332** is configured in two sections **332**A, **332**B, with the sections fitted within the through bore **22** across from one another. Each insert section **332**A, **332**B includes one or more voids or openings **334** passing through the insert body to permit fluid passage through the insert body, and therefore across the through bore **22**. As shown in the embodiment of FIG. **14**, the openings(s) **334** may be configured as cylindrical passages traversing through the insert **332** body. The insert **332** embodiment of FIG. **14** includes a plurality of openings **334** aligned in parallel with the longitudinal axis of the through bore **22**. Other insert **332** embodiments may be implemented with the openings **334** formed in different geometric configurations (e.g., oval, crescent, triangle, etc.) or with combinations of geometric shapes, all with the same dimensions or in varying dimensions. Some embodiments may be formed with the opening(s) **334** oriented or slanted with respect to the longitudinal axis of the through bore **22** (e.g., angled, spiral, or S-shaped channel). The openings **334** in the inserts **332** provide restriction channels that help diffuse and control fluid flow via the valve **10** through bore **22**, particularly in high-pressure fluid flow applications. The inserts **332** may be formed of any suitable material depending on the application (e.g., metal, composites, plastics, synthetic materials, etc.). The insert sections **332**A, **332**B may be easily removed and swapped to provide different diffusion effects by freeing an end cap **96** and removing the gate **26** as described herein.

[0052] As shown in FIG. **14**, some valve **10** embodiments may be configured with the gate **26** having one seal assembly **336**A, **336**B to provide sealing for both the gate opening **34** and the solid surface area **36** when the gate is in the closed position. FIG. **14** shows the valve **10** in the open position. The seal assembly **336**A, **336**B may be implemented with conventional radial seals to restrict fluid passage between the through bore **22** and the slot **28**, or with seals configured for energization to urge the respective seal faces against the surfaces to be sealed (e.g., as described in Intl. Pat. Apps. WO20211142004 and WO20211141999). Valve **10** embodiments of this disclosure may also be implemented with conventional or energizable radial/face seal carriers **340** disposed between the end caps **96** and main body **12** junctions.

[0053] The valve **10** embodiment of FIG. **14** is also implemented with an actuator **338** linked to the gate **26** to provide the motive force to move the gate along the transverse passage **28** between the open and closed positions. The actuator **338** is linked to one end of the gate **26**. Conventional actuators **338** may be used to implement valve **10** embodiments in accordance with this disclosure (e.g., EXLAR™ linear rotor screw actuators, hydraulic actuators, servos, etc.). Containment of the actuator **338** inside the valve **10** body **12** avoids the need for shaft seals, which prevents fugitive emissions (e.g., sand slurry and other substances entering the cavity between the gate and the body during service). Such embodiments provide a low-activation force and balanced gate valve **10**. When the valve **10** is to be opened, the actuator **338** is activated to apply a compressive or pushing force against the gate **26** (to the right in FIG. **14**). When the valve **10** is to be closed, the actuator **338** is activated to apply a tension or pulling force on the gate **26** (to the left in FIG. **14**). Power and/or fluid pressure for the actuator **338** may be provided via any suitable means as known in the art. Some embodiments may also be implemented with a wiper seal **342** disposed on the gate **26** near the end closest to the actuator **338**.

[0054] FIG. **15** shows an overhead view of the valve **10** of FIG. **14** with the gate **26** in the fully open position. In this mode, fluid flow via the through bore **22** is diffused by the plurality of openings **334** in the insert **332**. FIG. **16** shows an overhead view of the valve of FIG. **14** with the gate **26** in a near closed position. In this mode, fluid flow via the through bore **22** is restricted by the gate **26** and diffused by the uncovered openings **334** in the insert **332**. The actuator **338** may be

activated to move the gate **26** within the passage **28** to any desired position between fully closed and fully open.
[0055] An advantage of the disclosed valve **10** embodiments includes the implementation of a valve **10** that can be repaired/refurbished in the field, without needing to remove the valve unit from the operational system. Another advantage is the implementation of a valve **10** with the working components (e.g., seals) disposed in the gate **26**, making it very easy and efficient to replace key components via a swappable gate cartridge. The disclosed valves **10** do not require any packing or grease filling as used with conventional stemmed valve designs. By providing seals on both surfaces of the gate **26**, the valves **10** provide balanced upstream flow and downstream flow sealing between the gate and the body **12**. The disclosed valves **10** prevent fugitive emissions when the valve is in the open or closed position and while the gate **26** is in transition.
[0056] It will be appreciated that embodiments of the disclosed valves **10** may be implemented for use in numerous applications and operations, in the oil and gas industry and in other fields of endeavor. For example, the disclosed valve **10** embodiments may be deployed for use at surface, above surface, subsurface, and under water. It will be appreciated by those skilled in the art that embodiments of this disclosure may be implemented with conventional hardware components (e.g., conventional fasteners, seals, valve spools, etc.) and parts formed of suitable materials depending on the application. It will also be appreciated that embodiments may be implemented with control units locally or remotely linked to the valves **10** as known in the art. The control unit(s) may comprise any suitable microcomputer, processor, controllers, memory, and associated electronics, and may be programmed to activate and operate the valves **10** as described herein. In some embodiments, the control unit can be programmed to perform autonomous and automatic actuation of the valves and components as described herein. Power for the apparatus and valve **10** systems may also be implemented, for example, using conventional batteries as known in the art. Although only a few examples have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the examples. Accordingly, all such modifications are intended to be included within the scope of this disclosure.

Claims

1. A valve comprising: a main body having a through bore; a passage transverse to the through bore; a gate disposed in the transverse passage, wherein the gate is configured for motion between a position with an opening on the gate coincident with the through bore to permit fluid flow via the through bore and a position to restrict fluid flow via the through bore; and wherein the gate is configured for motion along the transverse passage in one direction in response to a first force acting on the gate and in another direction in response to a second force acting on the gate.
2. The valve of claim 1 further comprising an insert disposed in the main body coincident with the through bore; the insert configured with a plurality of openings formed thereon, wherein the gate is configured for movement along the transverse passage to control fluid flow via the plurality of openings.
3. The valve of claim 1 further comprising an actuator configured to apply a force on the gate to move the gate along the transverse passage.
4. The valve of claim 3 wherein the actuator is disposed inside the main body.
5. The valve of claim 1 wherein the gate comprises at least one seal to restrict fluid flow between the through bore and the transverse passage.
6. The valve of claim 1 wherein the gate comprises at least one seal configured for energization to restrict fluid flow between the through bore and the transverse passage.
7. The valve of claim 6 wherein the gate is configured with an internal fluid passage to channel fluid to energize the at least one seal configured for energization.
8. The valve of claim 1 wherein the body is configured with a first internal passage to channel fluid

to apply the first force on the gate to move the gate along the transverse passage.

9. The valve of claim 8 wherein the body is configured with a second internal passage to channel fluid to apply the second force on the gate in opposition to the first force to move the gate along the transverse passage.

10. The valve of claim 9 wherein the gate is configured for motion toward the position to restrict fluid flow via the through bore when one of the first force or the second force applied on the gate is reduced.

11. The valve of claim 8 wherein the body is configured with a second internal passage to channel a gas to apply the second force on the gate in opposition to the first force to move the gate along the transverse passage.

12. The valve of claim 11 wherein the gate is configured for motion toward the position to restrict fluid flow via the through bore when the second force is greater than the first force applied on the gate.

13. The valve of claim 1 wherein the gate has a first surface and a second surface opposite the first surface, with a first seal disposed on the first surface and a second seal disposed on the second surface.

14. The valve of claim 13 wherein the first seal and the second seal are each configured to restrict fluid flow between the through bore and the transverse passage when the gate is in the position to restrict fluid flow via the through bore.

15. The valve of claim 13 wherein the first seal and the second seal are each configured to restrict fluid flow between the through bore and the transverse passage when the gate is in the position to permit fluid flow via the through bore.

16. The valve of claim 1 wherein the gate is configured for motion free of any mechanical linkage.

17. The valve of claim 1 further comprising a cap mounted on the main body and configured for detachment to respectively permit removal and insertion of the gate from/into the body.

18. A method of operating a valve having a main body with a through bore, comprising: applying a force on a gate to move the gate along a passage in the main body transverse to the through bore, wherein the gate is configured for motion along the transverse passage between a position with an opening on the gate coincident with the through bore to permit fluid flow via the through bore and a position to restrict fluid flow via the through bore; and wherein the gate is configured for motion along the transverse passage in one direction in response to a first force acting on the gate and in another direction in response to a second force acting on the gate.